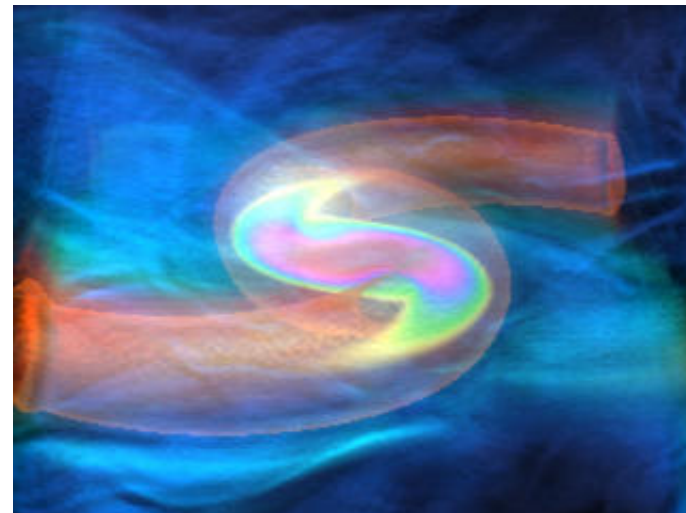
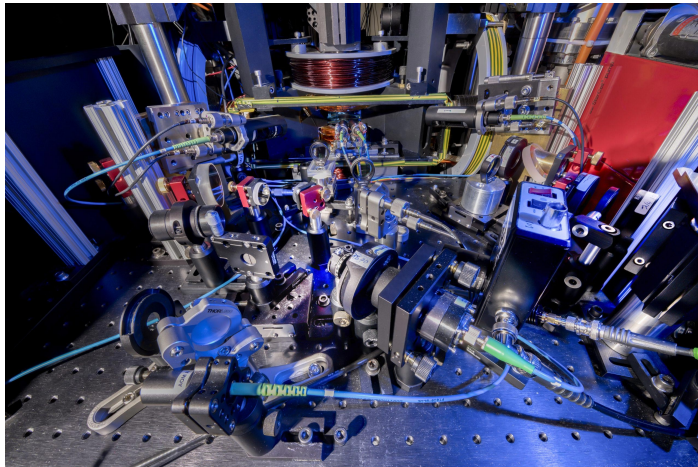


# The Unitary Fermi Gas as a Neutron Matter Emulator for Nuclear Astrophysics

Thomas Schaefer, North Carolina State University



## Why study the unitary Fermi gas?

Nuclear forces are quite complicated, and so is nuclear many-body theory (for the equation of state, reaction rates, mean free paths, transport properties, etc.).

The unitary Fermi gas is a very simple and clean model system for strongly correlated quantum many-body systems.

Universality connects atomic and nuclear systems.

Impressive progress in experimental control: Tune interactions, temperature, external fields, linear response, etc.

## What do we learn?

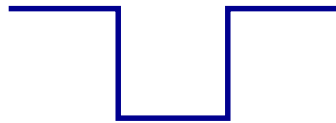
Numerical predictions that directly translate to nuclear systems (e.g. EOS).

Numerical benchmarks for nuclear many-body theory (EOS, response functions, reaction theory).

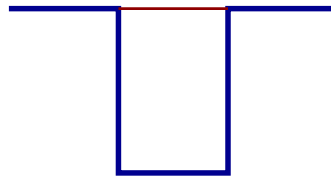
Improved understanding of few and many-body effects (short range correlations, reaction theory).

## Non-relativistic fermions in unitarity limit

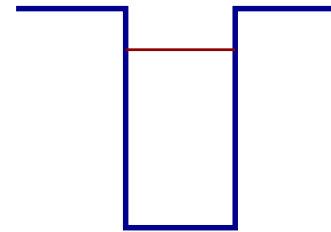
Two body interaction: Consider simple square well potential



$$a < 0$$



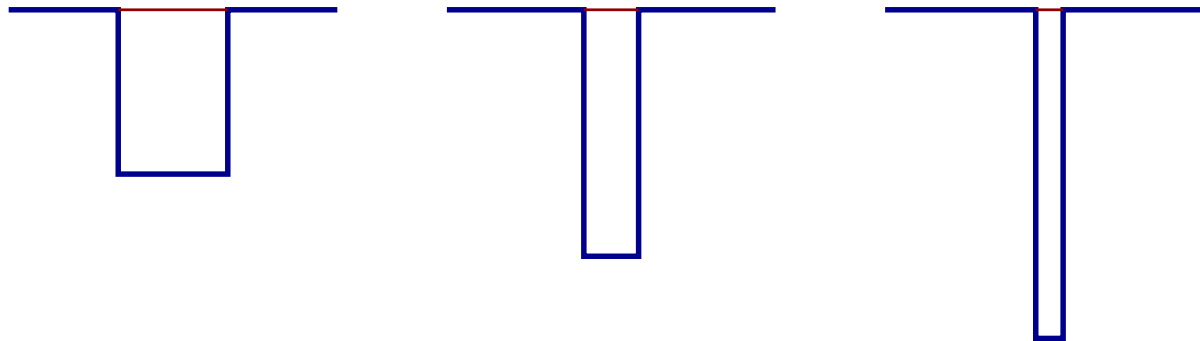
$$a = \infty, \epsilon_B = 0$$



$$a > 0, \epsilon_B > 0$$

## Non-relativistic fermions in unitarity limit

Now take the range to zero, keeping  $\epsilon_B \simeq 0$



$$\mathcal{T} = \frac{1}{ik}$$

The “Unitarity” Limit

## Fermi gas at unitarity: Field Theory

Non-relativistic fermions at low momentum

$$\mathcal{L}_{\text{eff}} = \psi^\dagger \left( i\partial_0 + \frac{\nabla^2}{2M} \right) \psi - \frac{C_0}{2} (\psi^\dagger \psi)^2$$

Unitarity limit:  $a \rightarrow \infty$  (DR:  $C_0 \rightarrow \infty$ )

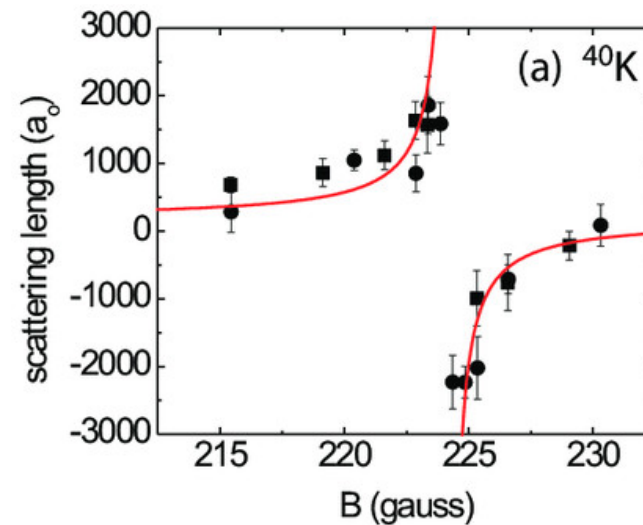
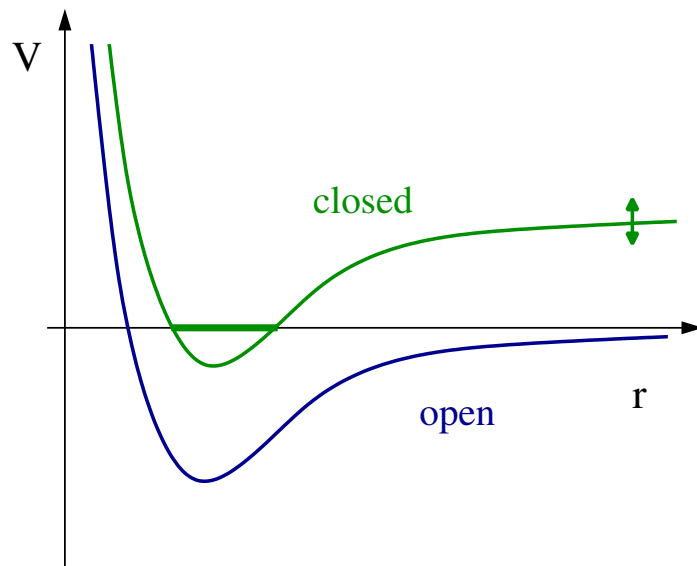
This limit is smooth: HS-trafo, define  $\Psi = (\psi_\uparrow, \psi_\downarrow^\dagger)$

$$\mathcal{L} = \Psi^\dagger \left[ i\partial_0 + \sigma_3 \frac{\vec{\nabla}^2}{2m} \right] \Psi + \left( \Psi^\dagger \sigma_+ \Psi \phi + h.c. \right) - \frac{1}{C_0} \phi^* \phi ,$$

$\phi \sim C_0 \psi_\uparrow \psi_\downarrow$  auxiliary “pair” or “dimer” field.

## Experimental realization: Feshbach resonances

Atomic gas with two spin states: “↑” and “↓”

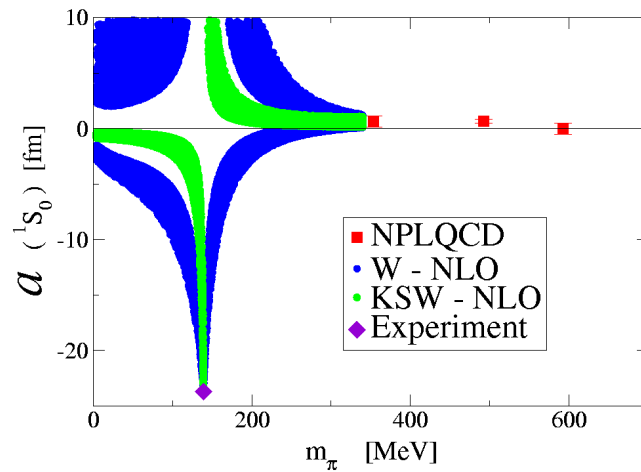


Feshbach resonance

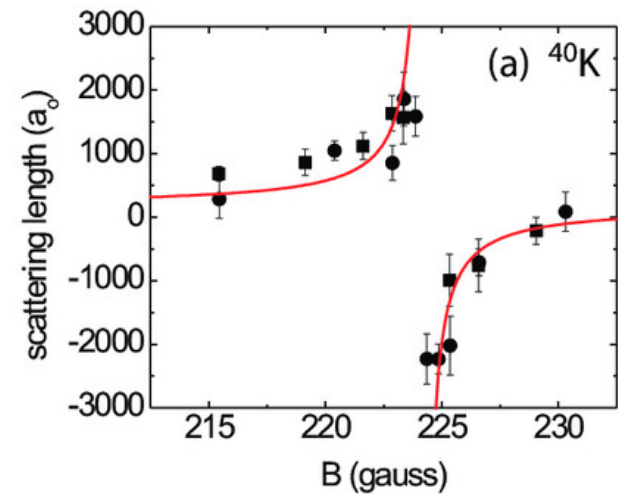
$$a(B) = a_0 \left( 1 + \frac{\Delta}{B - B_0} \right)$$

# Universality: From neutrons to atoms

## Neutron Matter



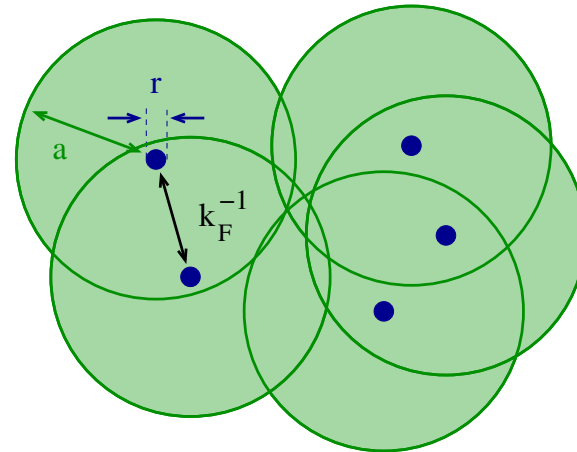
## $^{40}\text{K}$ Feshbach resonance



What do these systems have in common?

dilute:  $r\rho^{1/3} \ll 1$

strongly correlated:  $a\rho^{1/3} \gg 1$

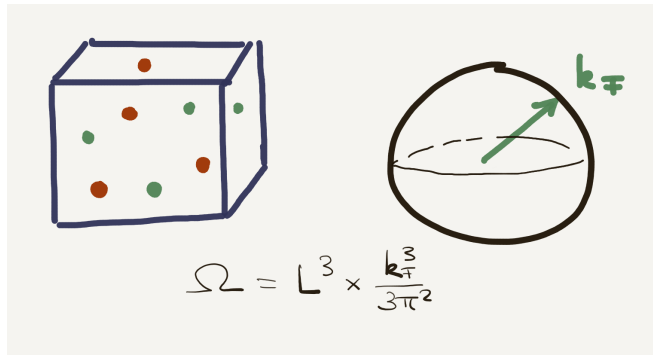


## Outline

1. **Equation of state:** From trapped atoms to neutron stars.
2. **The contact:** From the tail of the momentum distribution to short range correlations in nuclei.
3. **Mind the tail:** Response functions for  $\nu$ -scattering.
4. **Un-nuclear physics:** From trapped few-body systems to the disintegration of halo nuclei.
5. **Everything flows:** Elliptic flow from traps to heavy ions.

## 1. Equation of state

Free Fermi gas at zero temperature



$$\frac{E}{N} = \frac{3}{5} \frac{k_F^2}{2m} \quad \frac{N}{V} = \frac{k_F^3}{3\pi^2}$$

$$\mathcal{E} = E/V \sim (N/V)^{5/3}$$

Unitarity limit ( $a \rightarrow \infty$ ,  $r \rightarrow 0$ ). No dimensionful parameters.

$$\frac{E}{N} = \xi \frac{3}{5} \frac{k_F^2}{2m}$$

$$k_F \equiv (3\pi^2 N/V)^{1/3}$$

Prize problem (George Bertsch, 1998): Determine  $\xi$ .

Is  $\xi > 0$  (is the system stable)?

## How to measure $\xi$ with trapped atoms

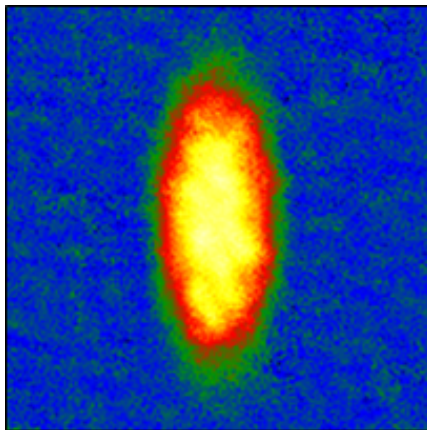
Trapped gas in hydrostatic equilibrium

$$\frac{1}{n} \vec{\nabla} P = -\vec{\nabla} V_{ext} \quad P = \frac{2}{3} \mathcal{E}$$

Pressure determines size of the cloud ( $V_{ext} = \frac{1}{2} m \omega^2 x^2$ ).

$$r(a=0) = \sqrt{\frac{2E_F}{m\omega^2}} \quad r(a=\infty) = \xi^{1/4} r(0)$$

Cloud size can be measured with a CCD camera and a ruler

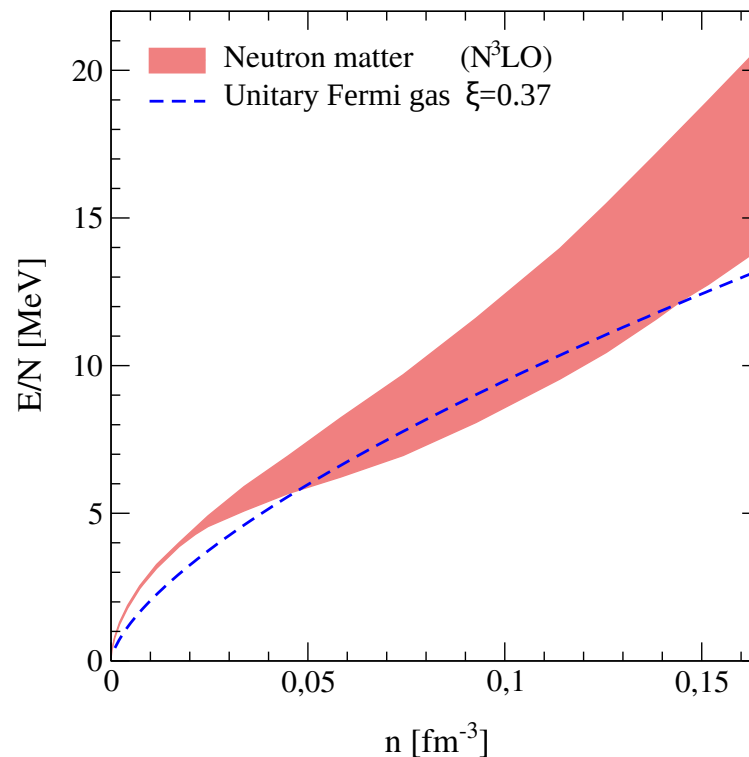


modern value

$$\xi = 0.376(5)$$

(MIT, Sommer et al.)

## Neutron matter equation of state

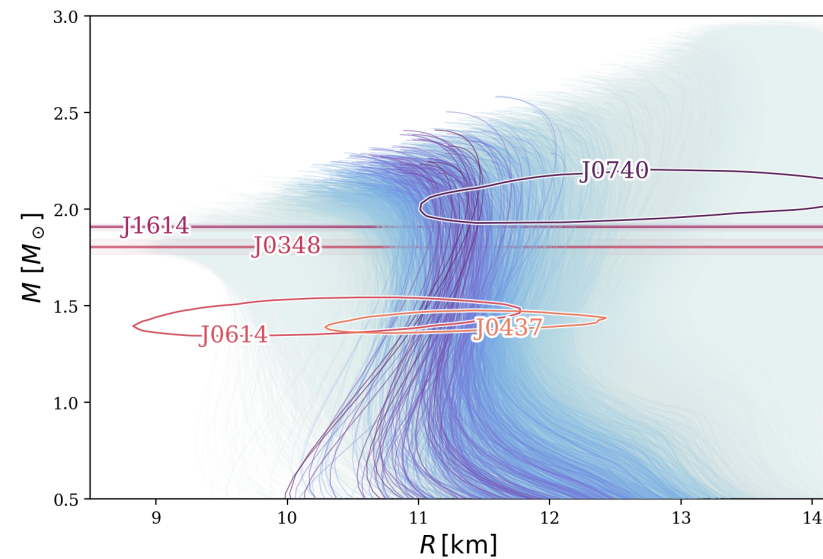
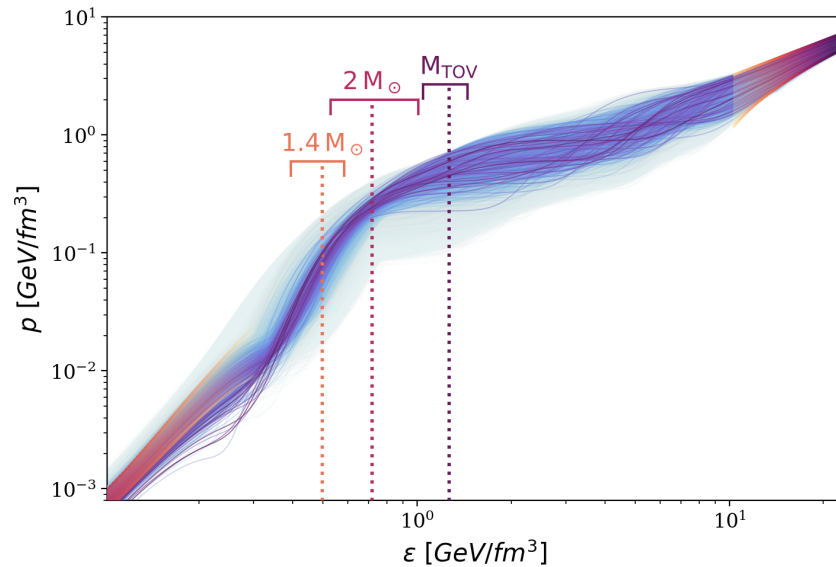


Schaefer, Schwenk  
Physik Journal (2014)

$n \lesssim 0.1 \text{ fm}^{-3}$ : Unitary gas with  $a^{-1}, r$  corrections.       $n \gtrsim 0.1 \text{ fm}^{-3}$ : Repulsive 2-body, 3-body forces.

$n \gtrsim 0.2 \text{ fm}^{-3}$ : New degrees of freedom.

## Neutron Matter EOS and Mass-Radius relation



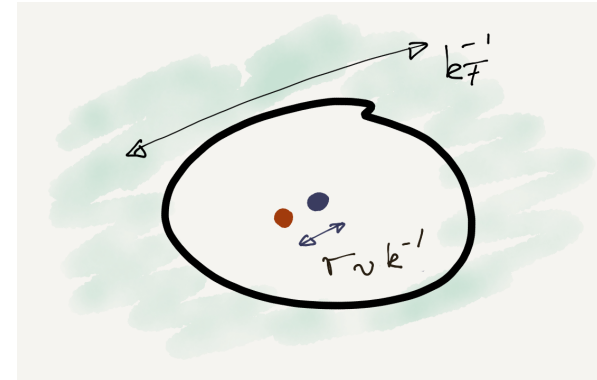
Use low (nuclear) and high (pQCD) density EOS as priors.  
Constrain intermediate densities by observation (NICER etc.).

## 2. Short range correlations and the “Contact”

Consider short distance structure of unitary gas

$$\langle n_{\uparrow}(R + r/2)n_{\downarrow}(R - r/2) \rangle \simeq \frac{1}{16\pi^2} \frac{\mathcal{C}}{r^2}$$

$\mathcal{C}$  : *Tan's contact density*



Related object: Momentum distribution

$$n_{\sigma}(k) = \int d^3R \int d^3r e^{-ikr} \langle \psi_{\sigma}^{\dagger}(R - r/2) \psi_{\sigma}(R + r/2) \rangle$$

The large momentum (short distance) tail of the distribution is

$$n_{\sigma}(k) = \frac{C}{k^4} \qquad C = \int d^3r \mathcal{C}(r) \quad \textit{Contact}$$

## Short range correlations, continued

The contact is related to the the pair density

$$\mathcal{C} = \langle m^2 \phi^\dagger \phi \rangle \quad \phi \sim C_0 \psi_\uparrow \psi_\downarrow$$

Many universal relations. Example: Thermodynamics

$$\left. \frac{dE}{da^{-1}} \right|_s = -\frac{h^2 \mathcal{C}}{4\pi m}$$

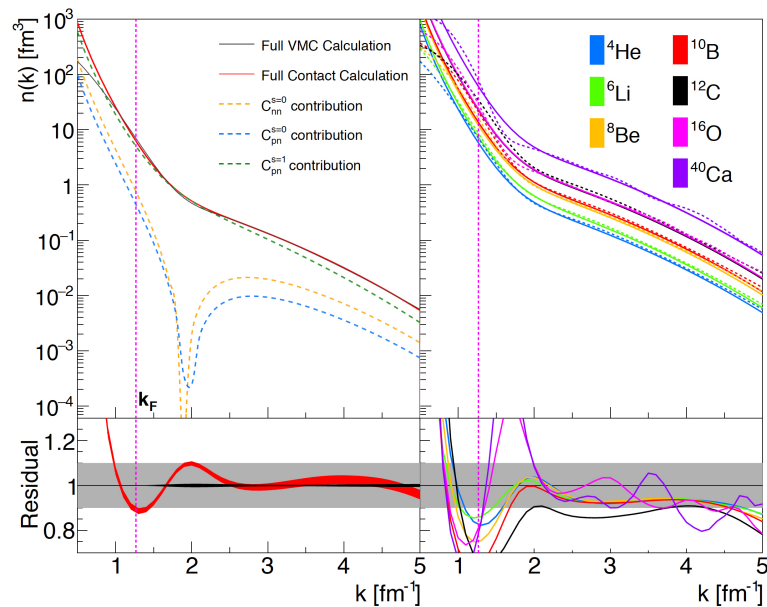
At high  $T$  the RHS can be computed using the Virial expansion

$$\mathcal{C} = \frac{8\pi^2 n^2}{mT}$$

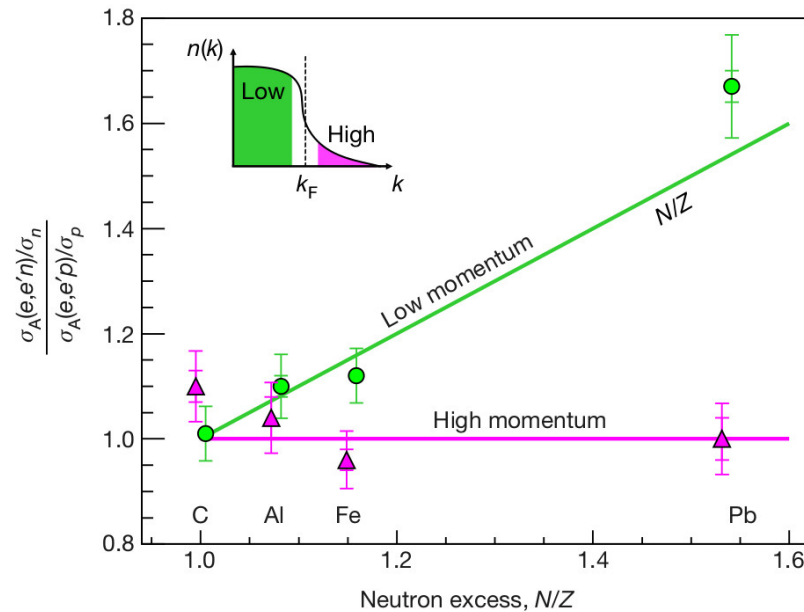
Momentum distribution  $n_\sigma \sim C/k^4$  fixed by EOS.

For  $T = 0$  have  $\mathcal{C} = \frac{2\zeta}{5\pi} (3\pi^2 n)^{4/3}$  with  $\zeta \simeq 0.95$ .

## Short range correlations and the contact in nuclei



Momentum distribution in  
nuclei (theory)  
Weiss et al. 1612.00923



Pair density in nuclei  
CLAS collaboration  
Nature (2018)

### 3. Structure Functions

Structure functions govern neutrino NC/CC current interactions

$$\frac{d^2\sigma_{NC}}{d\cos\theta d\omega} = G_F^2 [c_V f_V(\theta) S(\omega, q) + c_A f_A(\theta) S_\sigma(\omega, q)]$$

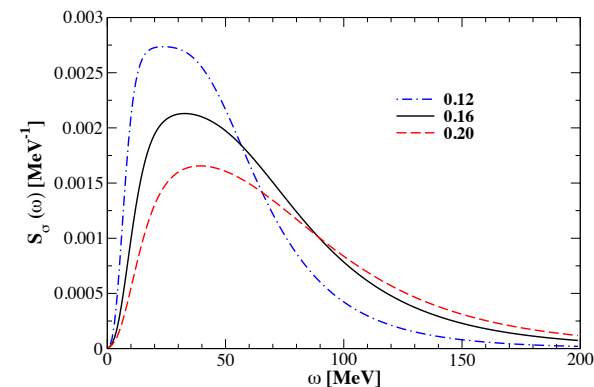
$$\frac{d^2\sigma_{CC}}{d\cos\theta dE_e} = G_F^2 [f_V(\theta) S_\tau(\omega, q) + g_A f_A(\theta) S_{\tau\sigma}(\omega, q)]$$

$$S_O = \int dt dx e^{i\omega t - iqx} \langle \rho_O(x, t) \rho_O(0, 0) \rangle, \quad \rho_A = \psi^\dagger O \psi,$$

Need response at fairly large frequency, e.g. energy loss rate

$$n + n \rightarrow n + n + \nu \bar{\nu}$$

$$Q \sim G_F^2 \int d\omega \omega^6 e^{-\omega/T} S_\sigma(\omega)$$



## Short range correlations

High frequency behavior (at fixed  $y = \omega/q^2$ ) controlled by OPE

$$A(x, t)B(0, 0) = \sum_i |x|^{\Delta_i - \Delta_A - \Delta_B} f_i(|x^2|/t) O_i(0)$$

Single particle response (RF spectroscopy)

$$I(\omega) = -\frac{1}{\pi} \text{Im} G(\omega) = \frac{\langle \phi^\dagger \phi \rangle}{4\pi^2} \frac{1}{\omega^{3/2}}$$

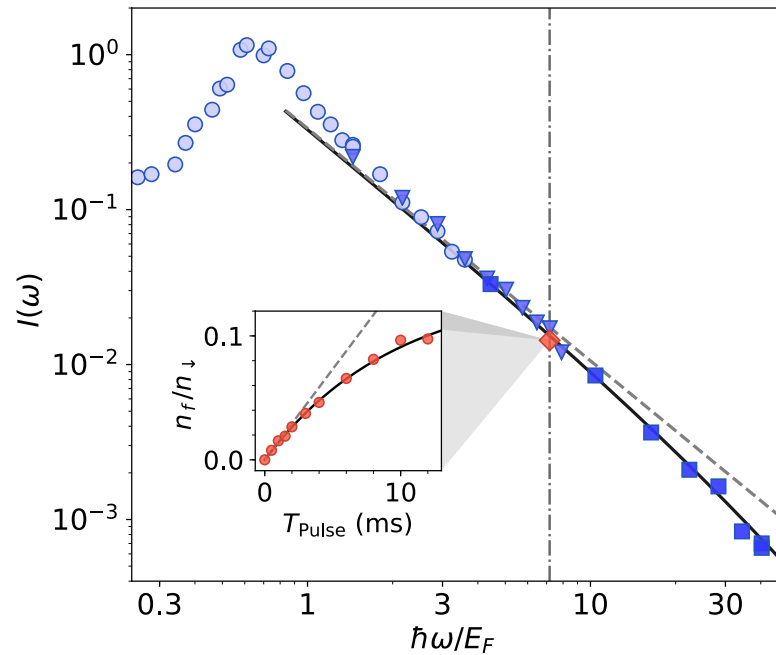
Structure factor (density)

$$S(\omega, q) = \frac{4\langle \phi^\dagger \phi \rangle}{45\pi^2 m} \frac{q^4}{\omega^{7/2}}$$

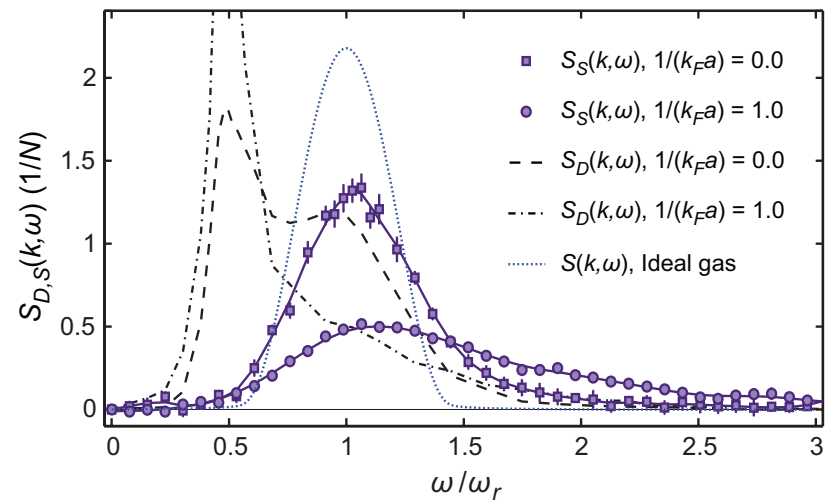
Structure factor (spin density)

$$S_\sigma(\omega, q) = \frac{\langle \phi^\dagger \phi \rangle}{3\pi^2 m^{1/2}} \frac{q^2}{\omega^{5/2}}$$

## Measurements of the response in cold gases



RF spectroscopy



Spin response

Mukherjee et al. (MIT), 2019. Vale et al. (Swinburne), 2012.

## 4. Conformal symmetry and Un-nuclear physics

Unitary Fermi gas is invariant under scale

$$x \rightarrow sx, \quad t \rightarrow s^2 t \quad [D, H] = 2iH \quad D = \int dx \, x \cdot j$$

and conformal transformations

$$x \rightarrow \frac{x}{1+ct}, \quad t \rightarrow \frac{t}{1+ct} \quad [C, H] = iD \quad C = \int dx \, \frac{x^2}{2} n$$

Constrains correlation functions, e.g. pair propagator

$$G_{\Phi}(\omega, p) = \frac{1}{\sqrt{p^2/(4m) - \omega}}$$

## Conformal symmetry: State operator correspondence

Generalized to operators  $\mathcal{U}_n(t, x)$  with higher mass  $M = nm$

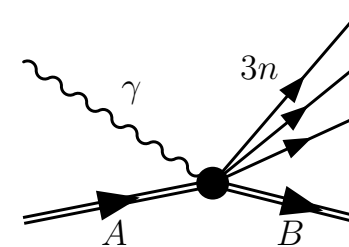
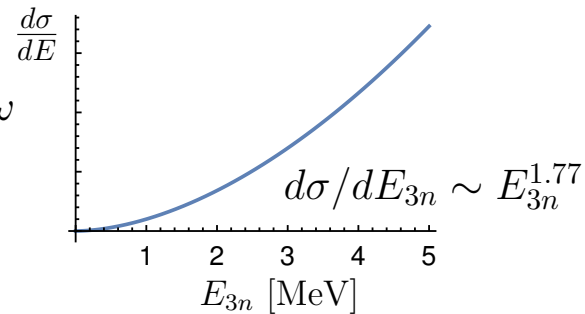
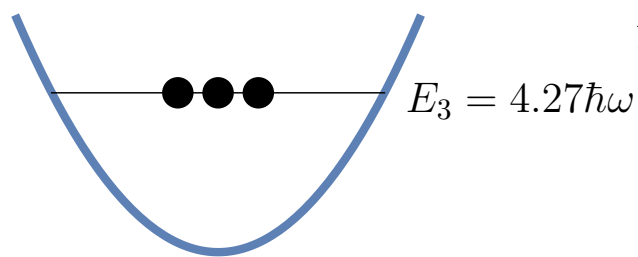
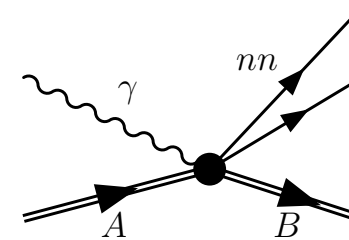
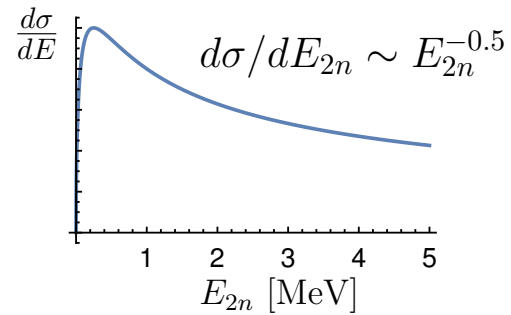
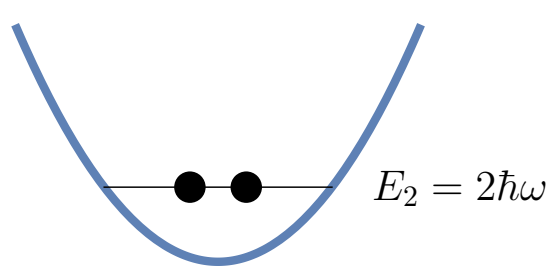
$$i\langle T\mathcal{U}_n^\dagger\mathcal{U}_n\rangle_{\omega,p} = \left(\frac{p^2}{2M} - \omega\right)^{\Delta_n-5/2} \quad \text{conformal dimension } \Delta_n$$

$\Delta_n$  related to ground state in harmonic potential

$$E_n = \Delta_n \hbar\omega \quad \text{state} - \text{operator correspondence}$$

E.g single free particle:  $\Delta_1 = 3/2$

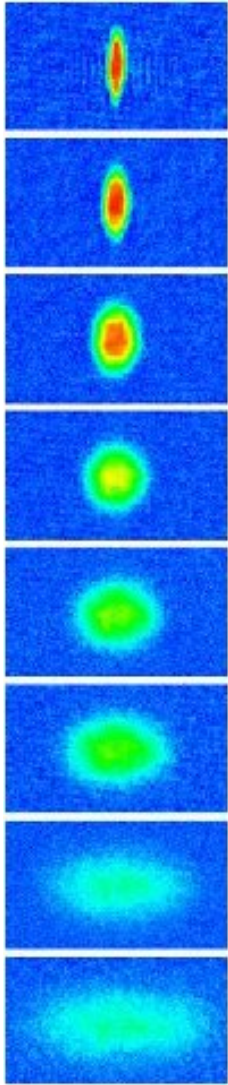
# Un-nuclear physics: Nuclear reactions



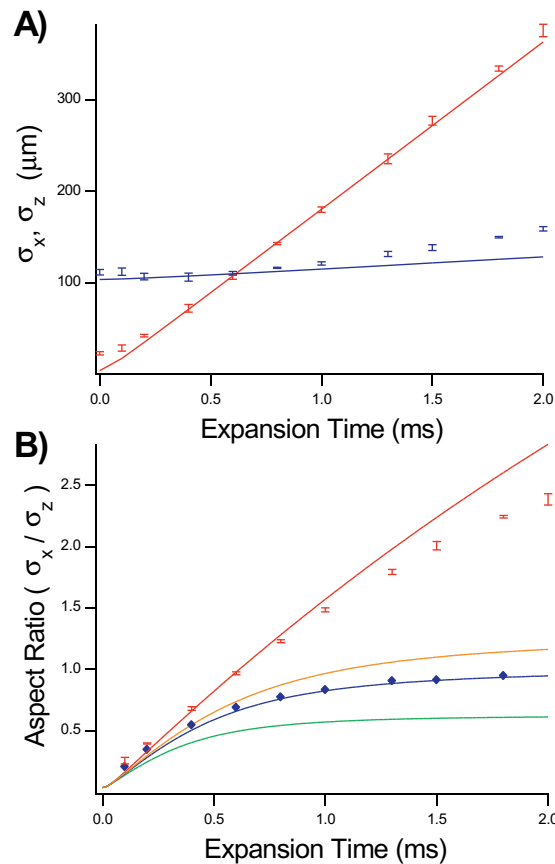
$$x + A \rightarrow B + Nn \quad \frac{d\sigma}{dE} \sim E^{\Delta_n - 5/2}$$

$$[(ma^2)^{-1} \sim 0.5 \text{ MeV}] < E < [(mr^2)^{-1} \sim 5 \text{ MeV}]$$

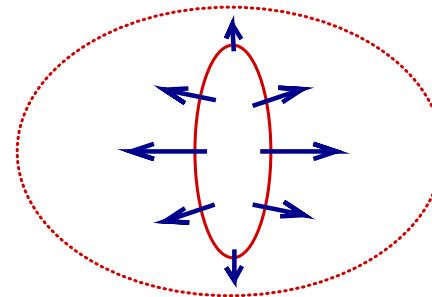
## 5. Elliptic flow in the unitary Fermi gas



O'Hara et al. (2002)

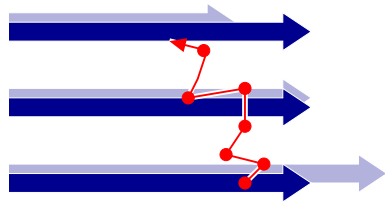


Hydrodynamic  
expansion converts  
coordinate space  
anisotropy  
to momentum space  
anisotropy



## Shear viscosity: Theory

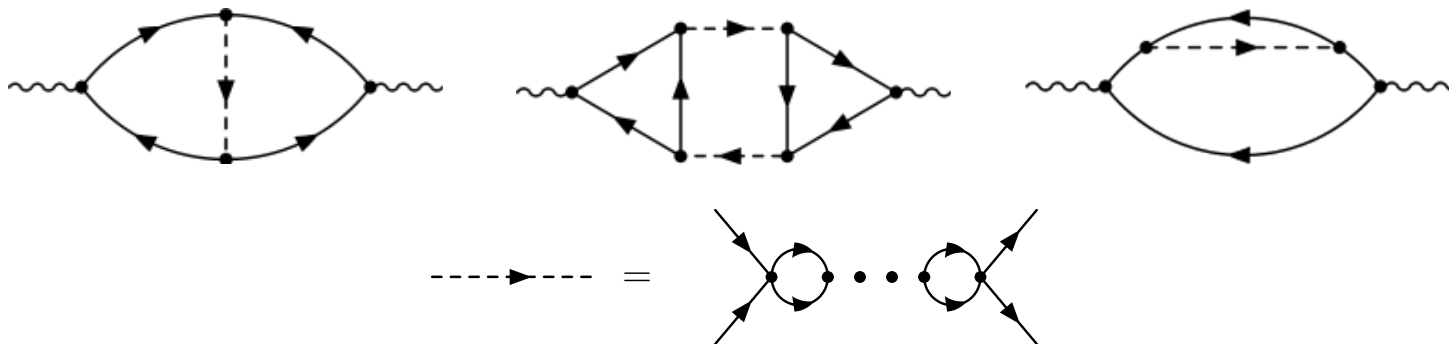
Kinetic theory: Momentum transport by diffusion of atoms



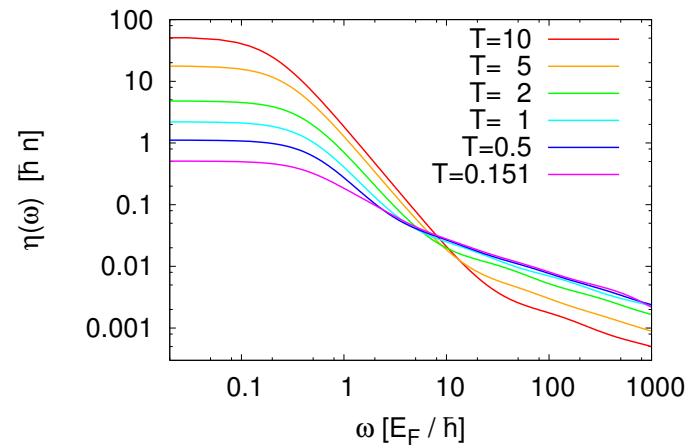
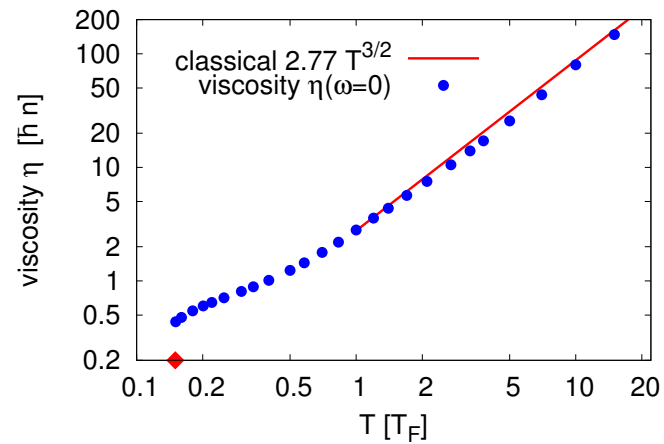
$$\eta = \frac{15}{32\sqrt{\pi}} (mT)^{3/2} \quad (T \gtrsim T_F)$$

QFT: Diagrammatic content of Boltzmann equation. Kubo formula with Maki-Thompson + Azlamov-Larkin + Self-energy

$$\eta = - \lim_{\omega \rightarrow 0} \frac{1}{\omega} \text{Im} \int dt d^3x e^{-i(\omega t - kx)} \Theta(t) \langle [\Pi_{xy}(0), \Pi_{xy}(t, x)] \rangle$$



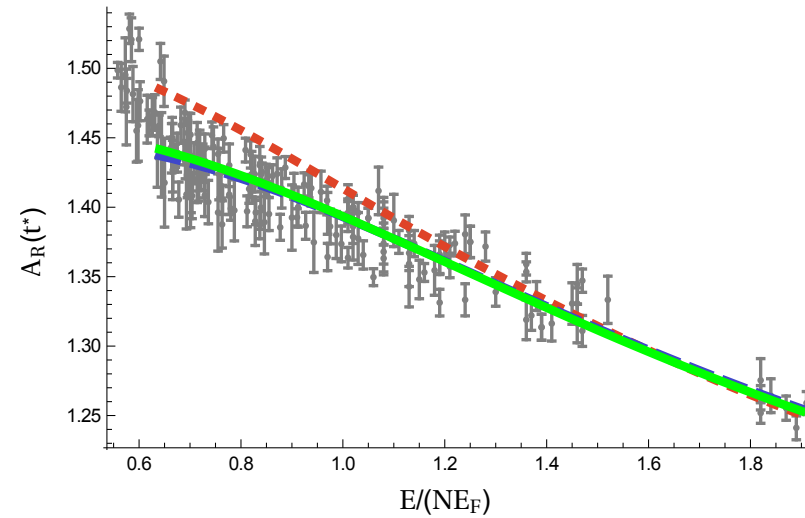
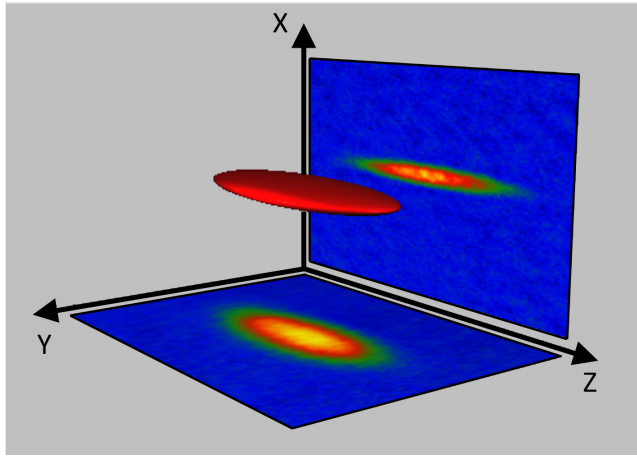
Can be used to extrapolate kinetic theory to  $T \sim T_F$



$$\eta(T \sim T_c) \sim \hbar n$$

Drude peak, universal tail.

## Shear viscosity: Phenomenology

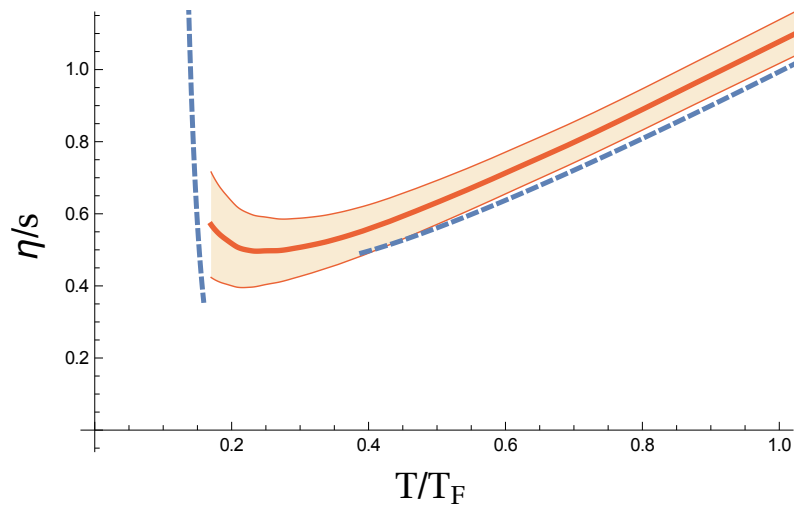


$A_R = \sigma_x / \sigma_y$  as function of total energy. Data: Joseph et al (2016).  $E/(NE_F) \sim 0.6$  is the superfluid transition.

Grey, Blue, Green: LO, NLO, NNLO fit.

$$\eta = \eta_0 (mT)^{3/2} \{1 + \eta_2 n \lambda^3 + \eta_3 (n \lambda^3)^2 + \dots\}$$

## Reconstruct $\eta/s$ (normal fluid)



$T_c \sim 0.17 T_F$ . Kinetic theory at low and high  $T$  (blue dashed)

Validate method at high temperature:

$$\eta|_{T \gg T_c} = (0.265 \pm 0.02)(mT)^{3/2}$$

$$\eta_0(th) = \frac{15}{32\sqrt{\pi}} = 0.269$$

Phenomenology: Two component model works well,  $\eta \sim \eta_0 T^{3/2} + \eta_1 n$

$$\eta/s|_{T_c} = 0.56 \pm 0.20$$

## Final thoughts

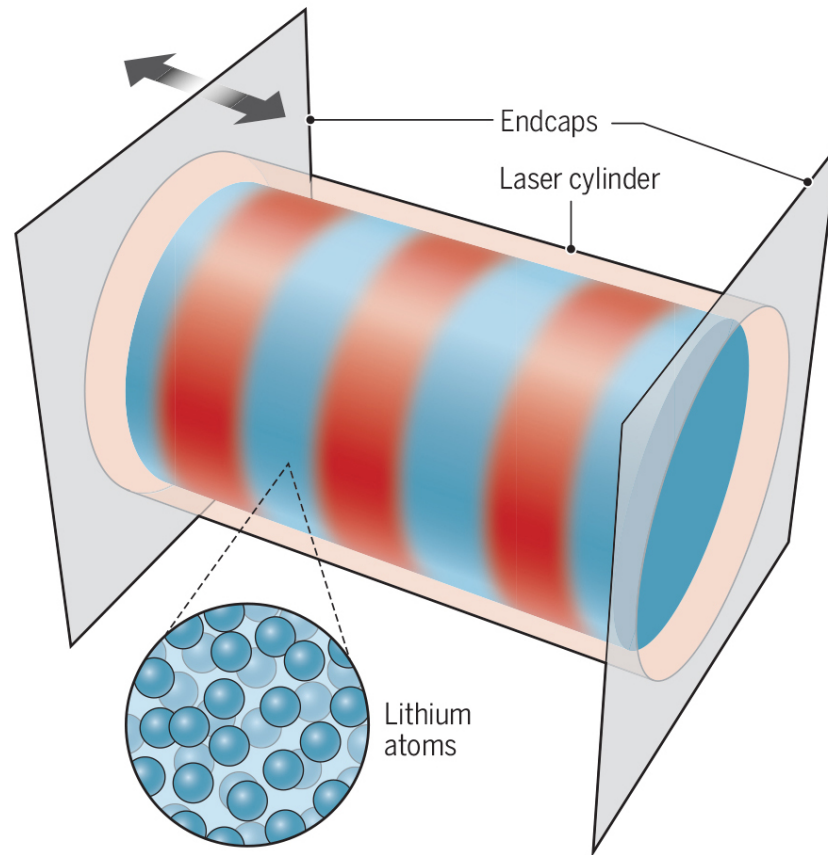
The unitary Fermi gas has become a paradigm for strongly correlated quantum liquids.

Universality relates the cold atomic gas to dilute neutron matter. Important for understanding analytic aspects, and as a benchmark for quantum Monte Carlo calculations.

Range of ideas continues to expand: From thermodynamics to transport, short range correlations and un-nuclear reactions.

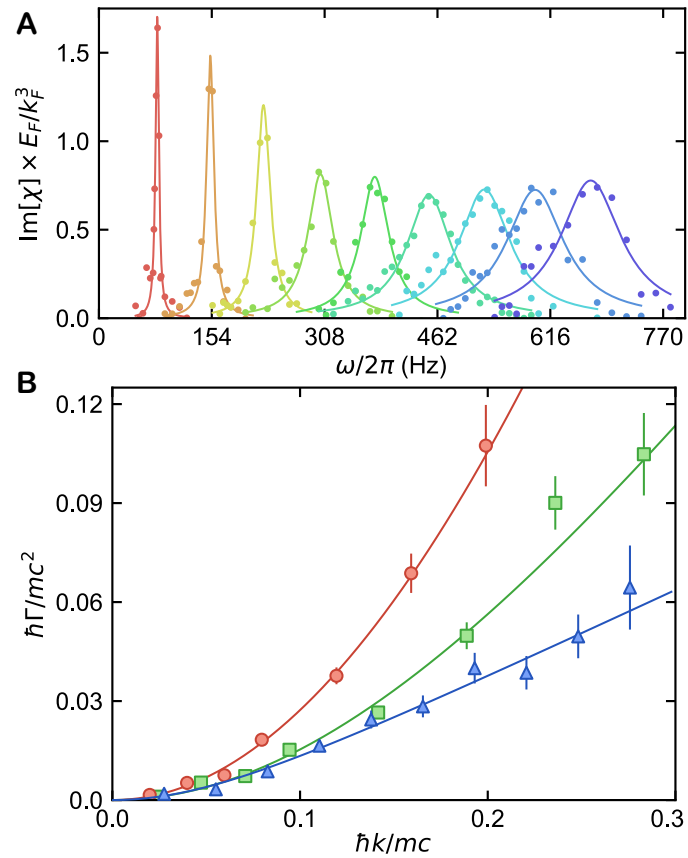
Continued role for ultracold gases as quantum simulators, not just universal gate based quantum computers.

## Sound attenuation (MIT)



# Sound attenuation (MIT)

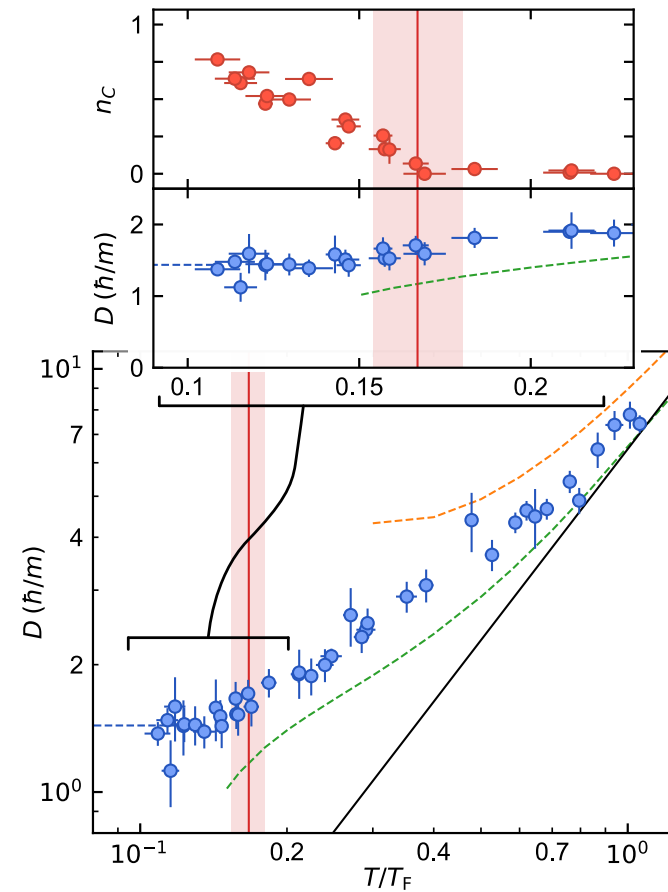
Spectral response  $\rho_k(\omega)$ .



Damping rate  $\Gamma(k)$

( $T/T_F = 0.36, 0.21, 0.13$ ).

Sound diffusivity  $D_s(T)$



$$D_s = \frac{4\eta}{3\rho} + \frac{4\kappa T}{15P}.$$

Patel et al., Science (2021)